

Ahmed Attia

personal website: copticoder.github.io

GitHub: github.com/Copticoder

ahmed.attia@mbzuai.ac.ae / ahmed.abdelsalam.ap@gmail.com

Mobile: +971 50 186 2157 / +20 1027 677 747

Location: Masdar City, Abu Dhabi, UAE

EDUCATION

- **Mohamed bin Zayed University of Artificial Intelligence** Abu Dhabi, United Arab Emirates
Master of Science in Natural Language Processing Aug 2024 - Present
- **Kafr El-Sheikh University Faculty of Artificial Intelligence** Kafr El-Sheikh, Egypt
Bachelor of Science in Artificial Intelligence; GPA: 3.7 Jul 2020 - Jul 2024

EXPERIENCE

- **Machine Learning Engineer, SpeedLegal** California, USA
Full-time Jul 2022 - Jun 2024
 - Collaborated in a team to build a Minimum Viable Product in 2 months for the goal of automating contract review process to help lawyers extract salient portions from contracts easily.
 - Developed Retrieval Augmented Generation pipelines with LLMs (**GPT-4**). Thereafter, properly evaluating the information retrieval system using precision@k, recall@k & mean reciprocal rank. This improved the contract review process by iterating and reaching a high recall@6 of 96% for NDA agreements.
 - Fine-tuned a pre-trained **BERT** model for an extractive question answering task on license agreements. The outcome was that the extractive model was able to do average of 20% exact match score and 65% on a set of 16 clauses leading to approximately 15 minutes decrease in contract review by the legal team.
 - Explored model Interpretability and Explainability (**LIME**). This helped in uncovering the details of the model reasoning in saying why it classified certain legal text under certain classes for reliability.
- **Undergraduate Teaching Assistant, Kafr El-Sheikh University** Kafr El-Sheikh, Egypt
Part-time Jan 2023 - Mar 2023
 - Assisted in teaching and preparing reinforcement learning lectures, materials as well as demonstrative lab exercises in the "Introduction to Multi-Agent Systems Design" course for second year students with the main instructor Fatma Nada.
 - Taught basic RL fundamentals including: Multi-Armed bandits, Markov Decision Processes (MDPs), Bellman Equations and optimality & Dynamic programming. (class size: 30 students)

SKILLS SUMMARY

- **Languages and Frameworks:**
C, Python, MongoDB, Unix scripting, Numpy, Pandas, Scikit-learn, PyTorch, Redis, Docker, GitLab CI, Streamlit, Design Patterns, Version Control, RESTful API, SOLID principles, Object Oriented Programming, Operating Systems, Agile Software Development, Reinforcement learning, Information retrieval.

ACADEMIC PROJECTS

- **DNG-MCTS— Reinforcement learning, Numpy, Scipy MBZUAI** Nov 2024
 - Implemented the paper: Bayesian mixture modelling and inference based Thompson sampling in Monte-Carlo tree search on **RaceTrack** environment.
 - Analyzed the performance of the DNG-MCTS variant against UCT MCTS, demonstrating faster convergence in **RaceTrack**.
 - Enhanced the algorithm's scalability and adaptability for TikZ program generation, integrating it with LLMs using the DeTikZify Vision-Language Model during inference.
- **Mengine: Meme retrieval engine using MemeCap dataset—NLTK, scikitlearn MBZUAI** Nov 2024
 - Developed and benchmarked multiple keyword-based retrieval models (**TF-IDF, Okapi BM25& The Binary Independence Model (BIM)**) for multi-modal meme retrieval in social media analysis, enhancing retrieval accuracy and relevance on a static dataset.
 - Preprocessed and prepared the MemeCap dataset, including OCR text extraction using **Microsoft/Florence-2-large** and using **LLaMA 3.2 90b** for generating structured query-document pairs for effective retrieval.
 - Evaluated retrieval performance with information retrieval metrics (AP@k, AR@k, MRR, MAP), providing a comprehensive analysis of model effectiveness in humor and context-driven tasks with a final best MRR score of 0.71 for the Binary Independence Model.
- **Crash PPO Agent— Reinforcement learning, Pytorch KFS University** Jan 2023
 - Implemented from scratch a policy gradient algorithm gave me insights and hands on experience on deep reinforcement learning, specifically, **Proximal Policy Optimization (PPO)** on an agent that plays the retro game Crash Bandicoot: the huge adventure.
 - Leveraged Pytorch for creating the actor and critic shared CNN architecture as well as the heads for the actor and the critic.
 - Leveraged openAI's gym retro environment. The agent learned to play one level of the game with an average cumulative rewards of 75+.

LANGUAGES

- **Arabic: Fluent, English: Excellent**

REFERENCES

References can be provided upon request.